GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

242490.1.pdf *Geochronology Record 1932*

242490.1: metamonzogranite, Wogara Bore

(Big Bell Suite, Austin Downs Supersuite; Youanmi Terrane, Yilgarn Craton)

SS Romano, MTD Wingate, Y LU, IOH Fielding and TJ Ivanic

Location and sampling

BELELE SG50-11, KALLI 2344 Zone 50, MGA 502693E 7031231N
Warox Site TJIKAL190115 Sampled on 2 September 2019

This sample was collected from bouldery outcrop on the north side of the Boolardy-Kalli Road on Kalli station, about 3.4 km west of 5 Mile Bore, 3.1 km southwest of Mudgeanoo Pool, and 2.8 km east of Wogara Bore.

Geological context

The unit sampled is a metamonzogranite (Fig. 1) of the 2732–2686 Ma Big Bell Suite of the Austin Downs Supersuite (Van Kranendonk et al., 2013). The Big Bell Suite consists mainly of porphyritic and equigranular metagranodiorite to metamonzogranite, which commonly exhibits a magmatic foliation (Zibra, 2021; ENS report Big Bell Suite, viewed 5 August 2023). A metamonzogranite sampled about 21 km to the west yielded an igneous crystallization age of 2701 ± 5 Ma (GSWA 242491, Romano et al., 2023). Several granitic samples collected about 37 km to the north yielded igneous crystallization ages of 2717–2680 Ma (e.g. GSWA 105017, Nelson, 1996). A tonalite gneiss collected about 91 km to the southeast yielded an igneous crystallization age of 2724 ± 2 Ma (GSWA 185923, Wingate et al., 2011).

Figure 1. Hand specimen image for sample 242490: metamonzogranite, Wogara Bore

Petrographic description

This sample is an inequigranular, fine to medium-grained metamonzogranite, consisting of about 35% plagioclase, 30% K-feldspar, 25% quartz, 7% biotite, 2–3% myrmekite, <1% magnetite and accessory allanite, titanite, zircon and apatite. Plagioclase is anhedral to subhedral, equant to slightly elongate, up to 2.5 mm long, and weakly saussuritized. K-feldspar is anhedral, irregular, microperthitic, up to 2 mm in size, and exhibits microcline twinning. Some K-feldspar grains contain abundant rounded quartz inclusions. Quartz is anhedral, irregular, strained, up to 1.5 mm in size, and interstitial to feldspar. The quartz–feldspar aggregate exhibits a semi-polygonal texture. Green–brown biotite is anhedral to subhedral, up to 1.5 mm in size, and partially replaced by chlorite. Myrmekite forms lobes up to 0.3 mm diameter around K-feldspar margins. Magnetite is anhedral to subhedral, equant, pitted and up to 0.5 mm in size. Allanite is euhedral, equant to slightly elongate, and up to 0.45 mm long, and titanite is anhedral and up to 0.3 mm in size. Apatite is euhedral and up to 0.2 mm long.

Although this rock is foliated in outcrop, the thin section lacks a penetrative tectonic fabric but contains abundant evidence of deformation and recovery at moderate metamorphic grade, in the form of microcline twinning, strain in quartz, myrmekite and semi-polygonal texture.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and subhedral to euhedral. The crystals are up to 400 µm long, and equant to elongate, with aspect ratios up to 6:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, inclusions of other minerals are abundant, and some crystals appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

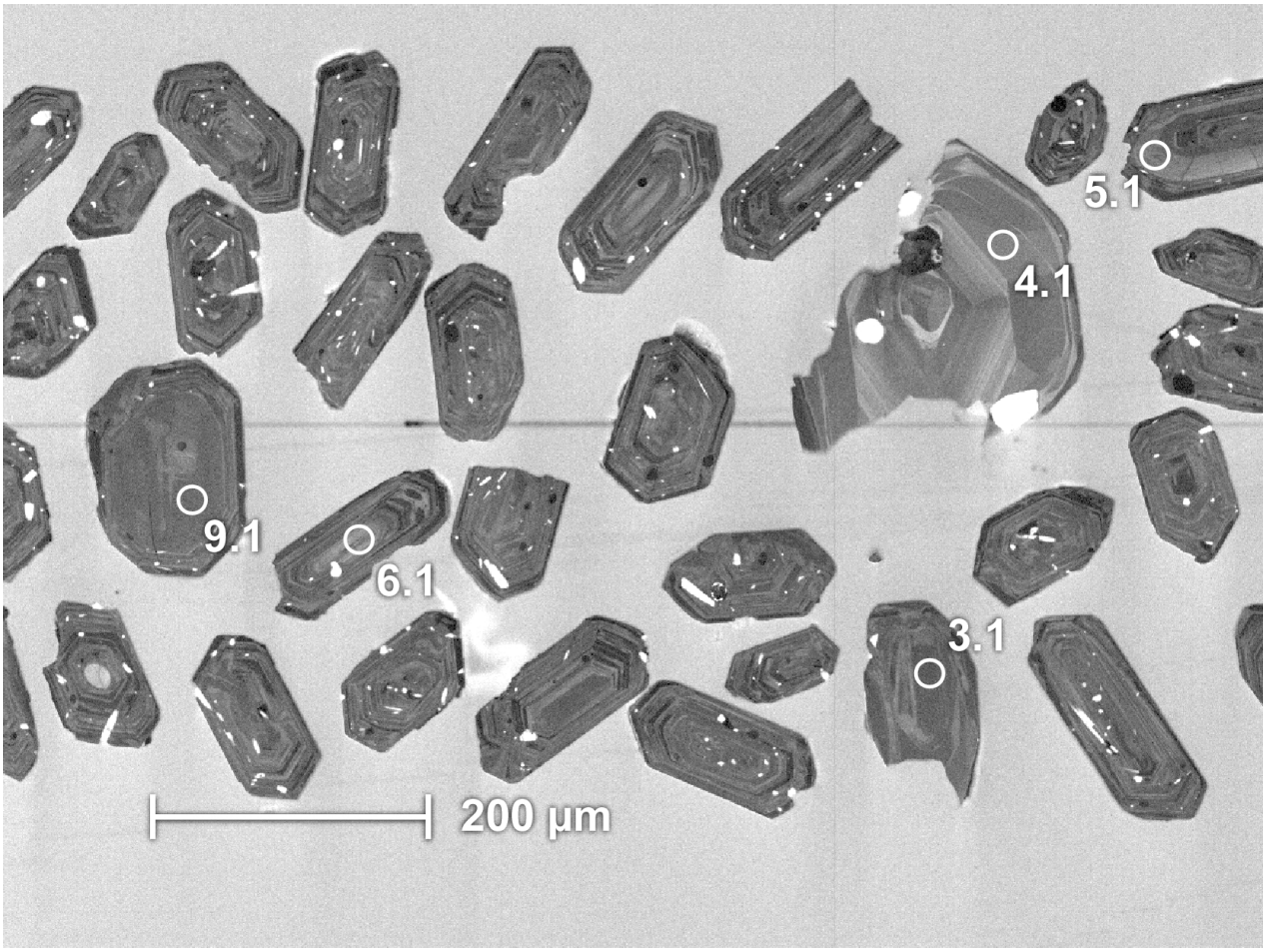

Figure 2. Cathodoluminescence image of representative zircons from sample 242490: metamonzogranite, Wogara Bore. Numbered circles indicate the approximate locations of analysis sites.

Analytical details

This sample was analysed on 11–12 February 2022, using SHRIMP-B. Nine analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.53% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.26% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Fourteen analyses were obtained from 14 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are slightly discordant to moderately discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable and is considered not to be geologically significant. The remaining 13 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ and Th/U ratios.

Group I comprises 11 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2693 ± 7 Ma (MSWD =

1.9). These analyses indicate mainly moderate to high Th/U ratios of 0.43 to 1.96 (median 1.16).

Group X comprises two analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2724 ± 8 Ma (MSWD = 0.062). These analyses indicate low Th/U ratios of 0.33 and 0.49.

The date of 2693 ± 7 Ma for the 11 analyses in Group I is interpreted as the magmatic crystallization age of the monzogranite. The date of 2724 ± 8 Ma for the two analyses in Group X is interpreted as the age of an inherited component.

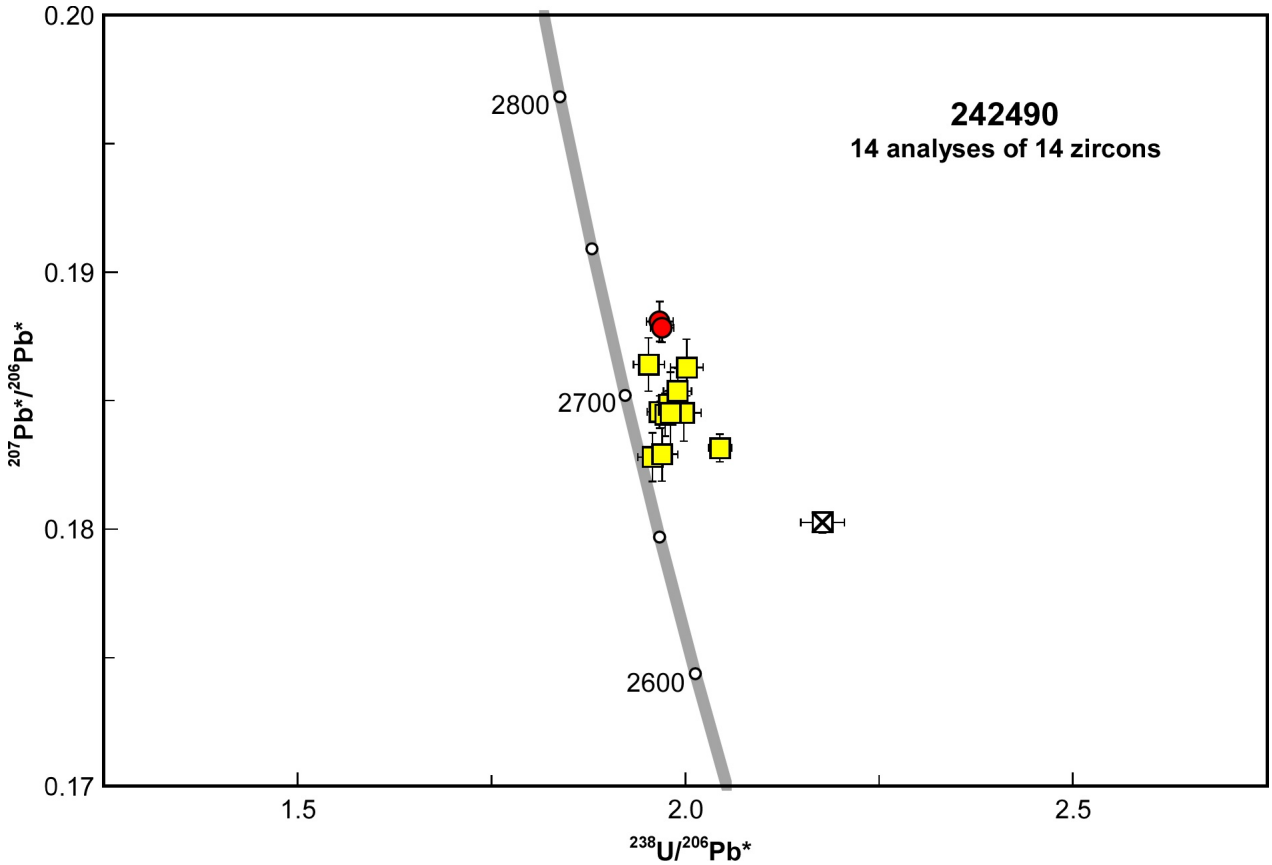

Figure 3. **U–Pb analytical data for zircons from sample 242490: metamonzogranite, Wogara Bore. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons); crossed square indicates Group D (discordance >5%)**

Links

Table 1: Ion microprobe analytical results for zircons from sample 242490: metamonzogranite, Wogara Bore: 242490.1.Table1.xls
SHRIMP spot images: 242490_spots.pdf
Introduction to geochronology information : Intro_2023.pdf

References

Nelson, DR 1996, 105017: fine-grained granite, Number 8 Bore 515: Geological Survey of Western Australia, 4p. View Reference online

Romano, SS, Wingate, MTD, Lu, Y, Fielding, IOH and Ivanic, TJ 2023, 242491.1: metamonzogranite, Piabubba Well; Geological Survey of Western Australia; Geochronology Record 1933: Geological Survey of Western Australia, <www.dmirs.wa.gov.au/geochron>.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

Wingate, MTD, Kirkland, CL and Van Kranendonk, MJ 2011, 185923: biotite–hornblende tonalite gneiss, Granite Rock 970: Geological Survey of Western Australia, 4p. View Reference online

Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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Acknowledgment of Country

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